

Tejasvi Konakanchi

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EDUCATION

University of Washington

Bachelor of Science in Computer Engineering

GPA: 3.61 / 4.0

Expected Graduation: 2027

Relevant Coursework: Digital Circuits & Systems, Microprocessor System Design, Electric Circuits, Continuous Time Linear Systems, Data Structures, Algorithms & Discrete Mathematics, Software Engineering, Technical Writing.

PROJECTS

8-bit Arithmetic Logic Unit (ALU) Design | Verilog, Quartus, DE10-Lite FPGA Mar. 2025 – Present

- Designed a high-performance 8-bit ALU supporting **16** arithmetic/logic operations by optimizing **RTL** code.
- Achieved **100%** functional coverage by developing a robust Verilog testbench with a **20**-case verification plan.
- Implemented a hardware-software interface on a Terasic DE10-Lite FPGA, mapping internal ALU states to onboard 7-segment displays and switches for real-time hardware validation.

Transient Response Analog Circuit | LTspice, Oscilloscope, Breadboarding May 2026 – Present

- Led a **3**-member team to design a second-order RLC circuit, analyzing transient step responses.
- Built a physical prototype on a breadboard and validated waveforms using function generators and oscilloscopes.
- Achieved **95%** alignment between physical hardware measurements and high-fidelity LTspice simulations.
- Authored a report reducing measurement discrepancies through rigorous statistical error analysis.

EXPERIENCE

Secretary Jun. 2026 – Present

Institute of Electrical and Electronics Engineers (IEEE)

Bothell, Washington

- Prepared meeting agendas while maintaining accurate notes to keep officers and members aligned on action items.
- Maintained chapter documentation using IEEE Vtools, ensuring compliance with IEEE reporting standards.
- Planned and led chapter events, driving member engagement by **15%** during peak periods.

Undergraduate Associate Jun. 2026 – Present

Sensors, Energy and Automation Laboratory (SEAL)

Seattle, Washington

- Developed Embedded-PPG documentation, improving accessibility by organizing **4** research projects.
- Established the team's communication workflow, improving coordination by defining collaboration channels.
- Aligned project documentation by coordinating with researchers and stakeholders, ensuring consistency.

Undergraduate Researcher Mar. 2026 – Jun. 2026

ECG Modeling – Continuous Time Linear Systems (B EE 235)

Bothell, Washington

- Designed a **3**-stage analog signal-conditioning circuit that recovered clean ECG waveforms from raw signals.
- Prototyped the full analog front-end on breadboard hardware using op-amps within lab material constraints.
- Collaborated in a **4**-person team to deliver a working ECG system and present it at the UW STEM Symposium.

Data Analyst Intern Jun. 2025 – Dec. 2025

Tesan IT Solutions

Hyderabad, India

- Streamlined management of **2,000+** student records by integrating databases, improving retrieval speed.
- Supported administrator portal operations by documenting workflows, improving team maintainability.

SKILLS

Programming Languages: C/C++, Verilog, Java, R, HTML/CSS, Python

Software & Tools: MATLAB, Quartus II, LTspice, Git/GitHub, VS Code, Eclipse, IntelliJ, R Studio, Figma

Testing & Lab: Oscilloscope, Multimeter, Logic Analyzer, Breadboarding, PCB Design

Soft Skills: Technical documentation, problem-solving, communication, teamwork, self-driven, adaptability

AWARDS & ACHIEVEMENTS

Dean's List Academic Award (2026): Recognized each quarter for high academic achievement on the Dean's List

President's Education Awards Program (2023): Honors K–12 students for academic excellence

Washington State Honor Award (2023): Awarded to the top 10% of Washington state high school graduates